Signal Model Analysis

Industry Benchmarking & Improvement Roadmap

Analysis of our current event-tracking to signal-summarization to recommendation pipeline against industry research and best practices. Goal: validate the model's merits, identify gaps, and design both an initial tuning framework and ongoing optimization strategy.

Part 1: What the Industry Tracks

Event Taxonomy (standard implicit signals)

Signal Type	Weight Class	Industry Tools	Our Coverage
Page view	Passive	GA4, Amplitude, Mixpanel, Segment	Yes (page_view)
Scroll depth (25/50/75/90%)	Passive	GA4 (90%), Hotjar, FullStory	No
Element visibility / dwell	Passive	Contentsquare, Mouseflow	Yes (view, 800ms+)
Hover / mouseover dwell	Passive	Mouseflow, custom	Yes (hover, 1000ms+)
Click	Active	All platforms	Yes (click)
Search query	Active (explicit)	Algolia, GA4 site search	Yes (search)
Tab/filter selection	Active	Custom builds	Yes (tab_view)
Add to cart / wishlist	Active (high)	Shopify, GA4 ecommerce	No
Form interaction	Active	Hotjar, FullStory	No
Scroll velocity / rage clicks	Frustration	FullStory, LogRocket	No
Session duration	Aggregate	GA4, Amplitude	No (implicit)
Return visit frequency	Loyalty	Amplitude, Mixpanel	Partial
Exit / bounce	Negative	GA4	No

Research-Backed Weight Hierarchies

Industry consensus weight ordering:

```
purchase (10x) > add-to-cart (5x) > click (3x) > hover/dwell (1.5-2x) > view (1x) > impression (0.5x)
```

Our current weights:

```
click_group (3x) > click_tag (2x) > hover_group/tab_group (2x) > hover_tag/tab_tag (1.5x) > view (1x)
```

Assessment: Our click and hover weights are within the industry range. The tag vs group distinction (click_tag=2 vs click_group=3) is a unique design choice not commonly seen in industry.

Part 2: Current Model — Strengths

1. **Recency weighting** — `0.5 + 0.5 * age` is a reasonable linear recency function. Research supports recency as one of the strongest predictors (Xie et al. 2022).
2. **Frequency bonus** — `log2(count)` is a sound sub-linear scaling choice. Prevents power users from overwhelming the model. Aligns with TF-IDF log-dampening.
3. **Cross-session exponential decay** — $0.8^{\text{sessionsAgo}}$ correctly models the forgetting curve. Research shows exponential decay outperforms linear.
4. **Hover-as-signal** — Tracking hover dwell aligns with research showing dwell time is a positive correlate of explicit ratings (Springer 2014).
5. **Score normalization** — `score / sqrt(tags.length)` prevents multi-tagged items from dominating. Similar to cosine normalization in TF-IDF systems.
6. **Diversity filter** — `maxPerGroup = 3` prevents recommendation homogeneity, a known problem in content-based systems.
7. **Client-side, zero-backend** — Privacy-preserving by design. Aligns with 2025-2026 privacy-first analytics trends (Consent Mode v2, cookieless tracking).

Part 3: Current Model — Gaps & Improvements

A. Event Capture Gaps

Gap	Impact	Recommendation
No scroll depth tracking	Missing key engagement signal	Add IntersectionObserver sentinels at 25/50/75/100%
No negative signals	Can't distinguish rejection from non-exposure	Track "viewed but not clicked" as weak negative
No dwell-weighted views	1s view = 30s view	Weight: min(dwellMs / 5000, 2.0)
View events ignored in scoring	Views carry no tags in summarizer	Ensure view events carry item tags

B. Weighting Model Improvements

Current	Problem	Improvement
Hardcoded weights	Can't tune without code changes	Make configurable via create() options
Tag vs group distinction	Unusual split not seen in industry	Simplify: weight by event type only
Linear recency (0.5+0.5*age)	Doesn't match research	Exponential: 0.3 + 0.7 * age^2
Frequency: click/hover only	Views never get bonus	Include views at 0.5x weight
Search: always 0.9 confidence	Vague = specific	Scale by query word count
Comparison: always 0.85	2 or 3 clicks identical	Scale: 0.7 + 0.15 * uniqueClicks

C. Decay Model Improvements

Current	Problem	Improvement
Within-session: 0.8/flush	Decays to 0.33 after 30s idle	Time-based: 0.8^(minutes/5)

Cross-session: 0.8^count	1hr vs 2wk gap = same decay	Time-based: 0.8^(days/3)
No session cap	Profile grows unbounded	Prune >30 days or cap at 20
No intent expiry	Inactive intents persist forever	Delete after 3+ sessions inactive

D. Pattern Detection Improvements

- **Only 6 signal categories** — Missing: browsing momentum, deep dive, return interest, fading interest
- **Group interest needs 3+ events** — Cold start problem. Lower to 2 for first session
- **Tag affinity 35% threshold** — Broad browsers never trigger it. Add diverse exploration signal
- **Hover interest: same group only** — Cross-group hover patterns invisible. Detect by shared tags
- **No abandoned interest** — After 2+ sessions without re-engagement, emit fading interest signal

E. Recommendation Engine Improvements

- **No IDF weighting** — Common tags (beach on 20/24 items) are non-discriminative. Apply: weight * log(totalItems / itemsWithTag)
- **No negative scoring** — Items viewed but never clicked across sessions should be penalized
- **Reason string: top tag only** — Show top 2-3 matched tags for richer explanations
- **No serendipity/exploration** — Filter bubble risk. Reserve 1-2 of 6 slots for exploration
- **Cold start is abrupt** — Show popularity-based defaults until 3+ events accumulate

Part 4: Tuning Framework

Phase 1 — Make the Model Configurable

Extract all hardcoded constants into a modelConfig object:

```
modelConfig: {
weights: { click: 3, hover: 2, view: 1, search: 2.5, tab_view: 1.5 },
recency: { min: 0.3, max: 1.0, curve: 'exponential' },
frequency: { minCount: 2, formula: 'log2' },
decay: { withinSession: 0.8, crossSession: 0.8, method: 'time-based' },
thresholds: { groupInterestMin: 3, tagAffinityPct: 0.35 },
recommendations: { maxResults: 6, maxPerGroup: 3, idfWeighting: true }
}
```

Phase 2 — Add Instrumentation

- Log signal-to-recommendation pipeline for each rec shown
- Track recommendation click-through rate (CTR) as positive feedback
- Tag each profile with the modelConfig version that produced it

Phase 3 — Iterative Tuning Loop

- **Baseline:** Run current model, measure rec CTR over N sessions
- **Weight sweep:** Try click=2/3/4, hover=1/1.5/2/3, measure CTR delta
- **Decay sweep:** Try 0.7/0.8/0.9 session decay, measure staleness vs freshness
- **Threshold sweep:** Try tagAffinity 25%/35%/50%, measure signal diversity
- **Compare:** Use debug panel or metrics page to visualize A vs B

Part 5: Priority Ranking

Priority	Improvement	Effort	Impact
P0	Make weights/thresholds configurable	Medium	High — enables all tuning
P0	Add IDF weighting to rec scoring	Low	High — discriminative tags
P1	Time-based cross-session decay	Low	Medium — accurate modeling
P1	Dwell-weighted view scoring	Low	Medium — graduated signal
P1	Cold start fallback (popularity)	Low	High — first-visit UX
P2	Scroll depth tracking	Medium	Medium — richer picture
P2	Negative signals (view-not-click)	Medium	Medium — reduces bad recs
P2	Exploration slots in recs	Low	Medium — reduces filter bubble
P3	Search confidence by specificity	Low	Low — marginal
P3	Session pruning (cap/expire)	Low	Low — storage hygiene
P3	Rec CTR tracking for tuning	Medium	High (long-term)

References

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- Beyond Explicit and Implicit: How Users Provide Feedback (2025)
- GA4 User Engagement Metrics — Root and Branch Group
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- Mouseflow — User Behavior Analysis Guide
- Product Engagement Score — Blitzllama
- Customer Engagement Score — Lifesight
- Content-Based Recommendations — TF-IDF and Cosine Similarity
- Top 7 Analytics Best Practices for 2026 — Trackingplan
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